

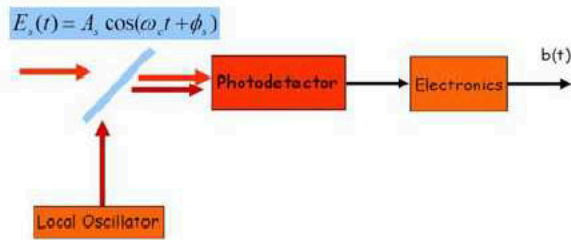
EE558 HW4

Problem 1:

- (a) The figure on the left corresponds to NRZ and the figure on the right corresponds to RZ. For the same bit rate, the pulses (i.e., time-domain) in RZ signal are narrower and thus have broader optical spectrum (i.e., frequency-domain).
 (b) NRZ signal is more tolerant to dispersion because of the narrow optical spectrum.
 (c) Peak (i.e., clock tone) spacing is equal to the bit rate, 10 GHz.

Problem 2:

(a) and (b)



$$P_{tot} = P_s + P_{LO} + 2\sqrt{P_s P_{LO}} \cos((\omega_c - \omega_{LO})t + \Delta\phi)$$

$$+ 2\sqrt{P_s P_{LO}} \cos((\omega_c + \omega_{LO})t + \phi_c + \phi_{LO})$$

High frequency component is not passed and $P_s \ll P_{LO}$

$$\approx P_{LO} + 2\sqrt{P_s P_{LO}} \cos(\omega_{IF}t + \Delta\phi)$$

$$I_{tot} = I_{DC} + I_{signal}(t) = R_{res} P_{tot}$$

$$= R_{res} P_{LO} + 2R_{res} \sqrt{P_s P_{LO}} \cos(\omega_{IF}t + \Delta\phi)$$

It is necessary to have $P_s \ll P_{LO}$ because if P_s and P_{LO} are comparable, we cannot neglect P_s in DC part of output current.

(c)

Increasing 3 dB means $P_{LO2} = 2P_{LO1}$. Thus:

$$I_{signal2}(t) = \sqrt{2} I_{signal1}(t) = 1.41 I_{signal1}(t)$$

$$I_{DC2} = 2 I_{DC1}$$

(d)

$$P_{s2}P_{LO2} = 10^{0.3-0.2}P_{s1}P_{LO1} = 1.26 \times P_{s1}P_{LO1}$$

Then,

$$I_{signal2}(t) = \sqrt{1.26}I_{signal1}(t) = 1.12 \times I_{signal1}(t)$$

$$I_{DC2} = 2 I_{DC1}$$

Problem 3:

$$L_{eff} = \frac{\Phi_{NL}}{\gamma P_{in}} = \frac{\Phi_{NL} A_{eff} \lambda}{2\pi n_2 \gamma P_{in}} = 748.9 \text{ km}$$

Problem 4:

(a) Because the distance of constellation points in 16-QAM is less than this distance in QPSK. Thus, the probability of error in 16-QAM is considerable higher than QPSK. On the other hand, each point carries 4 bits in 16-QAM whereas each point carries 2 bits in QPSK which means higher spectral efficiency.

(b) If noise is not the issue, so it is better to use 16-QAM because it would use less bandwidth.

(c) If bandwidth is not the issue, so QPSK would be better option because it is more tolerant.